Midsem Solutions

Solutions to Questions 1–6

Question 1 : Matrix Form

Linear regression in matrix form

$$Y = XW$$

$$W = \begin{bmatrix} a \\ b \end{bmatrix}$$

$$W = (X^T X)^{-1} X^T Y$$

Design Matrix

$$X = \begin{bmatrix} 1 & 7.04 \\ 1 & 8.77 \\ 1 & 9.52 \\ 1 & 12.28 \\ 1 & 12.36 \end{bmatrix}$$

$$Y = \begin{bmatrix} 13.49 \\ 16.89 \\ 18.50 \\ 23.80 \\ 23.99 \end{bmatrix}$$

Compute $X^T X$

$$X^T = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 7.04 & 8.77 & 9.52 & 12.28 & 12.36 \end{bmatrix}$$

$$X^T X = \begin{bmatrix} 5 & 49.97 \\ 49.97 & 520.63 \end{bmatrix}$$

Compute $(X^T X)^{-1}$

Determinant

$$|X^T X| = 5(520.63) - (49.97)^2$$

$$= 2633.15 - 2497$$

$$= 106.15$$

$$(X^T X)^{-1} = \frac{1}{136.15} \begin{bmatrix} 520.63 & -49.97 \\ -49.97 & 5 \end{bmatrix}$$

Compute $X^T Y$

$$X^T Y = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 7.04 & 8.77 & 9.52 & 12.28 & 12.36 \end{bmatrix} \begin{bmatrix} 13.49 \\ 16.89 \\ 18.50 \\ 23.80 \\ 23.99 \end{bmatrix}$$

$$= \begin{bmatrix} 96.67 \\ 1007.99 \end{bmatrix}$$

Final Parameters

$$W = (X^T X)^{-1} X^T Y$$

$$= \frac{1}{136.15} \begin{bmatrix} 520.63 & -49.97 \\ -49.97 & 5 \end{bmatrix} \begin{bmatrix} 96.67 \\ 1007.99 \end{bmatrix}$$

Result

$$a \approx -0.339$$

$$b \approx 1.968$$

Final regression model

$$y = -0.339 + 1.968x$$

Question 2: Bayesian Classification

- We are doing Bayesian classification with Gaussian Likelihood.
- For each class ω_j:

$$P(x|\omega_j) = \frac{1}{(2\pi)^{d/2}|C_j|^{1/2}} \exp\left(-\frac{1}{2}(x-\mu_j)^T C_j^{-1}(x-\mu_j)\right)$$

Where:

- x: Input vector of dimension d.
- m: Number of classes.
- μ_j: Mean vector of class j.
- C_j: Covariance matrix of class j.

Number of Floating-Point Parameters

For each class ω_j:

- **Mean Vector ($\mu_j \in \mathbb{R}^d$):** d parameters.
- **Covariance Matrix ($C_j \in \mathbb{R}^{d\times d}$):** Since it is symmetrical , there are $\frac{d(d+1)}{2}$ unique elements.
- **Prior Probability ($P(\omega_j)$):** 1 parameter.

Total Parameters:

- **Per Class:** $d + \frac{d(d+1)}{2} + 1$.
- **For m Classes:** $m\left(d + \frac{d(d+1)}{2} + 1\right)$.

Minimum Number of Operations

Main computation is the Mahalanobis distance: $(x - \mu_j)^T C_j^{-1}(x - \mu_j)$.
Operations per class:

1. **Compute** $(x - \mu_j)$**:** d additions.
2. **Matrix-vector multiplication** $C_j^{-1}(x - \mu_j)$**:** d^2 multiplications and $d(d-1)$ additions.
3. **Dot product:** d multiplications and $d - 1$ additions.
4. **Scaling (-1/2):** 1 multiplication.
5. **Exponential evaluation:** 1 operation.
6. **Division (normalization):** 1 operation.

Summary of Computational Results

Total Operations for m Classes:

Operation Type	**Total for m Classes**
Multiplications	$m(d^2 + d + 1)$
Additions	$m(d^2 + d - 1)$
Divisions	m
Exponentials	m
Comparisons	$m - 1$

Question 3

Given

$$g(x; \theta) = \lambda \theta^{x-1}$$

Normalization

$$\sum_{x=1}^{\infty} g(x; \theta) = 1$$

Finding λ

$$\sum_{x=1}^{\infty} \lambda \theta^{x-1} = 1$$

$$\lambda \sum_{x=0}^{\infty} \theta^{x} = 1$$

Geometric series

$$\sum_{x=0}^{\infty} \theta^{x} = \frac{1}{1-\theta}$$

$$\lambda = 1 - \theta$$

Likelihood

Sample

$$S = \{x_1, x_2, \ldots, x_n\}$$

Likelihood

$$L(\theta) = \prod (1-\theta)\theta^{x_i - 1}$$

$$L(\theta) = (1-\theta)^n \theta^{\sum(x_i - 1)}$$

Log Likelihood

$$\ln L = n \ln(1-\theta) + \left(\sum x_i - n\right) \ln \theta$$

Derivative

$$\frac{d}{d\theta} \ln L = -\frac{n}{1-\theta} + \frac{\sum x_i - n}{\theta}$$

MLE

Setting derivative to zero

$$-\frac{n}{1-\theta}+\frac{\sum x_i - n}{\theta}=0$$

Result

$$\theta=\frac{\sum x_i - n}{\sum x_i}$$

$$\theta=1-\frac{n}{\sum x_i}$$

- $S_1 = \{x_i^{(1)}\} \quad i = 1, 2, \ldots, n_1$
- $S_2 = \{x_j^{(2)}\} \quad j = 1, 2, \ldots, n_2$

$$z_i^{(1)} = p^T x_i^{(1)} \qquad z_j^{(2)} = p^T x_j^{(2)}$$

- $\bar{x}^{(1)} = \frac{1}{n_1} \sum_{i=1}^{n_1} x_i^{(1)} = mean = m_1$
- $\bar{x}^{(2)} = \frac{1}{n_2} \sum_{j=1}^{n_2} x_j^{(2)} = mean = m_2$

$$(m_1 - m_2)^2 = \left[p^T \left(\frac{1}{n_1} \sum_{i=1}^{n_1} x_i^{(1)} - \frac{1}{n_2} \sum_{j=1}^{n_2} x_j^{(2)} \right) p \right]^2$$

Variance and Scatter Matrices

$$\sigma_1^2 = \sum_{i=1}^{n_1}(p^T x_i^{(1)} - m_1)^2 = p^T \left[\sum_{i=1}^{n_1}(x_i^{(1)} - \bar{x}^{(1)})(x_i^{(1)} - \bar{x}^{(1)})^T\right] p$$

$$\sigma_2^2 = \sum_{j=1}^{n_2}(p^T x_j^{(2)} - m_2)^2 = p^T \left[\sum_{j=1}^{n_2}(x_j^{(2)} - \bar{x}^{(2)})(x_j^{(2)} - \bar{x}^{(2)})^T\right] p$$

within class matrix

$$S_W = \sum_{i=1}^{n_1}(x_i^{(1)} - \bar{x}^{(1)})(x_i^{(1)} - \bar{x}^{(1)})^T + \sum_{j=1}^{n_2}(x_j^{(2)} - \bar{x}^{(2)})(x_j^{(2)} - \bar{x}^{(2)})^T$$

Projected variance becomes $p^T S_W p$

Optimization and Solution

LDA maximizes the ratio $\frac{(m_1 - m_2)^2}{\sigma_1^2 + \sigma_2^2}$

$$J(p) = \frac{p^T(\bar{x}^{(1)} - \bar{x}^{(2)})^2}{p^T S_W p}$$

$$p \propto S_W^{-1}(\bar{x}^{(1)} - \bar{x}^{(2)})$$

$$\hat{p} = \frac{S_W^{-1}(\bar{x}^{(1)} - \bar{x}^{(2)})}{\|S_W^{-1}(\bar{x}^{(1)} - \bar{x}^{(2)})\|}$$

Question 5

Quadratic logistic regression

$$\log \frac{y}{1-y} = x^T A x + b^T x + c$$

Discriminant

$$g(x) = x^T A x + b^T x + c$$

Feature Expansion

Let

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Quadratic expansion

$$g(x) = a_{11}x_1^2 + 2a_{12}x_1x_2 + a_{22}x_2^2 + b_1x_1 + b_2x_2 + c$$

Transformation

Define

$$\phi(x) = \begin{bmatrix} x_1^2 \\ x_2^2 \\ x_1 x_2 \\ x_1 \\ x_2 \\ 1 \end{bmatrix}$$

Then

$$g(x) = w^T \phi(x)$$

Thus quadratic logistic regression becomes linear in expanded feature space.

Question 5: Quadratic Logistic Regression

a) Model Definition:

- Dataset: $S = \{(x_i, y_i), i = 1 \ldots n\}$, where $y_i \in \{0, 1\}$ and $x_i \in \mathbb{R}^d$
- Quadratic Logit function:

$$z(x) = x^T A x + b^T x + c = \log \frac{p(x)}{1 - p(x)}$$

- Solving for $p(x)$:

$$\Rightarrow \frac{p(x)}{1 - p(x)} = e^{z(x)} \implies p(x) = \frac{1}{1 + e^{-z(x)}}$$

Question 5

Resulting class probabilities:

- $P(y = 1|x) = \frac{1}{1+\exp(-(x^T A x + b^T x + c))} = \hat{y}$
- & $P(y = 0|x) = 1 - P(y = 1|x)$

b) Likelihood Functions:

- Likelihood function $L(A, b, c) = \prod_{i=1}^{n} (\hat{y}_i)^{y_i} (1 - \hat{y}_i)^{1-y_i}$ where $\hat{y}_i = \sigma(z_i)$
- Log-likelihood $\ell(A, b, c) = \sum_{i=1}^{n} [y_i \log \hat{y}_i + (1 - y_i) \log(1 - \hat{y}_i)]$

Gradient Derivation:

- The partial derivative with respect to parameters θ:

$$\frac{\partial \ell}{\partial \theta} = \sum_{i=1}^{n} (y_i - \hat{y}_i) \frac{\delta z_i}{\delta \theta}$$

Parameter Update Rules

So, the update rules are defined as follows:

Parameter	**Update Rule**
c	$c^{t+1} \leftarrow c^t + \eta \sum_{i=1}^{n} (y_i - \hat{y}_i)$
b	$b^{t+1} \leftarrow b^t + \eta \sum_{i=1}^{n} (y_i - \hat{y}_i) x_i$
A	$A^{t+1} \leftarrow A^t + \eta \sum_{i=1}^{n} (y_i - \hat{y}_i) x_i x_i^T$

Step 1: Given Data Sets

We are provided with two distinct data sets, S_1 and S_2:

- **Set** S_1: $\{-3.27, 5.11, -1.13, 1.21, 3.22\}$
- **Set** S_2: $\{7.93, 9.13, 8.79, 13.19, 12.01\}$

Calculated Parameters:

- **Means:** $\mu_1 = 1.028, \quad \mu_2 = 10.21$
- **Variances:** $\sigma_1^2 = 11.16, \quad \sigma_2^2 = 5.13$

Step 2: Formula and Substitution

The Mahalanobis distance between these two sets is calculated using the following formula:

Formula

$$D = \sqrt{\frac{|\mu_1 - \mu_2|^2}{\frac{\sigma_1^2+\sigma_2^2}{2}}}$$

Substituting our values into the equation:

$$D = \sqrt{\frac{|1.028 - 10.21|^2}{\frac{11.16+5.13}{2}}}$$

Step 3: Final Calculation

Simplifying the terms inside the square root:

1. **Numerator (Squared Difference):** $|1.028 - 10.21|^2 \approx 84.31$
2. **Denominator (Average Variance):** $\frac{11.16+5.13}{2} = \frac{16.29}{2}$

Solving for D:

$$D = \sqrt{\frac{84.31}{16.29} \times 2}$$

Result

$$\mathbf{D = 3.217}$$